

Brownian Motion

Brownian motion is a process of tremendous practical and theoretical significance. It originated (a) as a model of the phenomenon observed by Robert Brown in 1828 that “pollen grains suspended in water perform a continual swarming motion,” and (b) in Bachelier’s (1900) work as a model of the stock market. These are just two of many systems that Brownian motion has been used to model. On the theoretical side, Brownian motion is a Gaussian Markov process with stationary independent increments. It lies in the intersection of three important classes of processes and is a fundamental example in each theory.

The first part of this chapter develops properties of Brownian motion. In Section 7.1 we define Brownian motion and investigate continuity properties of its paths. In Section 7.2 we prove the Markov property and a related 0-1 law. In Section 7.3 we define stopping times and prove the strong Markov property. In Section 7.4, we use the strong Markov property to investigate properties of the paths of Brownian motion. In Section 7.5 we introduce some martingales associated with Brownian motion and use them to obtain information about exit distribution and times. In Section 7.6 we prove three versions of Itô’s formula.

7.1 Definition and Construction

A one-dimensional **Brownian motion** is a real-valued process B_t , $t \geq 0$ that has the following properties:

- (a) If $t_0 < t_1 < \dots < t_n$, then $B(t_0), B(t_1) - B(t_0), \dots, B(t_n) - B(t_{n-1})$ are independent.
- (b) If $s, t \geq 0$, then

$$P(B(s+t) - B(s) \in A) = \int_A (2\pi t)^{-1/2} \exp(-x^2/2t) dx$$

- (c) With probability one, $t \rightarrow B_t$ is continuous.

(a) says that B_t has independent increments. (b) says that the increment $B(s+t) - B(s)$ has a normal distribution with mean 0 and variance t . (c) is self-explanatory.

Thinking of Brown’s pollen grain (c) is certainly reasonable. (a) and (b) can be justified by noting that the movement of the pollen grain is due to the net effect of the bombardment of millions of water molecules, so by the central limit theorem, the displacement in any one interval should have a normal distribution, and the displacements in two disjoint intervals should be independent.

Two immediate consequences of the definition that will be useful many times are:

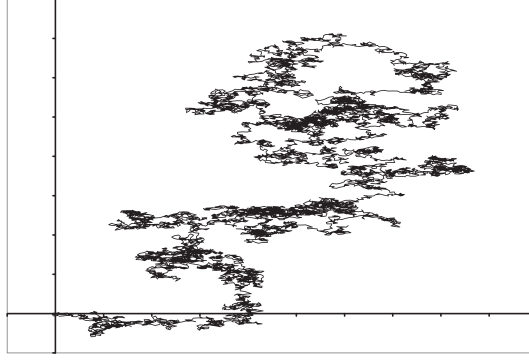


Figure 7.1 Simulation of two dimensional Brownian motion.

Translation invariance $\{B_t - B_0, t \geq 0\}$ is independent of B_0 and has the same distribution as a Brownian motion with $B_0 = 0$.

Proof Let $\mathcal{A}_1 = \sigma(B_0)$ and $\mathcal{A}_2$ be the events of the form

$$\{B(t_1) - B(t_0) \in A_1, \dots, B(t_n) - B(t_{n-1}) \in A_n\}.$$

The $\mathcal{A}_i$ are π -systems that are independent, so the desired result follows from the $\pi - \lambda$ theorem 2.1.6. $\square$

The Brownian scaling relation If $B_0 = 0$, then for any $t > 0$,

$$\{B_{st}, s \geq 0\} \stackrel{d}{=} \{t^{1/2} B_s, s \geq 0\} \quad (7.1.1)$$

To be precise, the two families of r.v.'s have the same finite dimensional distributions, i.e., if $s_1 < \dots < s_n$, then

$$(B_{s_1 t}, \dots, B_{s_n t}) \stackrel{d}{=} (t^{1/2} B_{s_1}, \dots, t^{1/2} B_{s_n})$$

Proof To check this when $n = 1$, we note that $t^{1/2}$ times a normal with mean 0 and variance s is a normal with mean 0 and variance st . The result for $n > 1$ follows from independent increments. $\square$

A second equivalent definition of Brownian motion starting from $B_0 = 0$, that we will occasionally find useful is that $B_t, t \geq 0$, is a real-valued process satisfying

(a') $B(t)$ is a **Gaussian process** (i.e., all its finite dimensional distributions are multivariate normal).

(b') $EB_s = 0$ and $EB_s B_t = s \wedge t$.

(c') With probability one, $t \rightarrow B_t$ is continuous.

It is easy to see that (a) and (b) imply (a'). To get (b') from (a) and (b), suppose $s < t$ and write

$$EB_s B_t = E(B_s^2) + E(B_s(B_t - B_s)) = s$$

The converse is even easier. (a') and (b') specify the finite dimensional distributions of B_t , which by the last calculation must agree with the ones defined in (a) and (b).

The first question that must be addressed in any treatment of Brownian motion is, "Is there a process with these properties?" The answer is "Yes," of course, or this chapter would not exist. For pedagogical reasons, we will pursue an approach that leads to a dead end and then retreat a little to rectify the difficulty. Fix an $x \in \mathbf{R}$ and for each $0 < t_1 < \dots < t_n$, define a measure on $\mathbf{R}^n$ by

$$\mu_{x,t_1,\dots,t_n}(A_1 \times \dots \times A_n) = \int_{A_1} dx_1 \cdots \int_{A_n} dx_n \prod_{m=1}^n p_{t_m - t_{m-1}}(x_{m-1}, x_m) \quad (7.1.2)$$

where $A_i \in \mathcal{R}$, $x_0 = x$, $t_0 = 0$, and

$$p_t(a, b) = (2\pi t)^{-1/2} \exp(-(b - a)^2/2t)$$

From this formula, it is easy to see that for fixed x the family μ is a consistent set of finite dimensional distributions (f.d.d.'s), that is, if $\{s_1, \dots, s_{n-1}\} \subset \{t_1, \dots, t_n\}$ and $t_j \notin \{s_1, \dots, s_{n-1}\}$, then

$$\mu_{x,s_1,\dots,s_{n-1}}(A_1 \times \dots \times A_{n-1}) = \mu_{x,t_1,\dots,t_n}(A_1 \times \dots \times A_{j-1} \times \mathbf{R} \times A_j \times \dots \times A_{n-1})$$

This is clear when $j = n$. To check the equality when $1 \leq j < n$, it is enough to show that

$$\int p_{t_j - t_{j-1}}(x, y) p_{t_{j+1} - t_j}(y, z) dy = p_{t_{j+1} - t_{j-1}}(x, z)$$

By translation invariance, we can without loss of generality assume $x = 0$, but all this says is that the sum of independent normals with mean 0 and variances $t_j - t_{j-1}$ and $t_{j+1} - t_j$ has a normal distribution with mean 0 and variance $t_{j+1} - t_{j-1}$.

With the consistency of f.d.d.'s verified, we get our first construction of Brownian motion:

Theorem 7.1.1 *Let $\Omega_o = \{\text{functions } \omega : [0, \infty) \rightarrow \mathbf{R}\}$ and $\mathcal{F}_o$ be the σ -field generated by the finite dimensional sets $\{\omega : \omega(t_i) \in A_i \text{ for } 1 \leq i \leq n\}$, where $A_i \in \mathcal{R}$. For each $x \in \mathbf{R}$, there is a unique probability measure ν_x on $(\Omega_o, \mathcal{F}_o)$ so that $\nu_x\{\omega : \omega(0) = x\} = 1$ and when $0 < t_1 < \dots < t_n$*

$$\nu_x\{\omega : \omega(t_i) \in A_i\} = \mu_{x,t_1,\dots,t_n}(A_1 \times \dots \times A_n) \quad (7.1.3)$$

This follows from a generalization of Kolmogorov's extension theorem, (7.1) in Appendix 3. We will not bother with the details, since at this point we are at the dead end referred to earlier. If $C = \{\omega : t \rightarrow \omega(t) \text{ is continuous}\}$, then $C \notin \mathcal{F}_o$, that is, C is not a measurable set. The easiest way of proving $C \notin \mathcal{F}_o$ is to do Exercise 7.1.4.

This problem is easy to solve. Let $\mathbf{Q}_2 = \{m2^{-n} : m, n \geq 0\}$ be the **dyadic rationals**. If $\Omega_q = \{\omega : \mathbf{Q}_2 \rightarrow \mathbf{R}\}$ and $\mathcal{F}_q$ is the σ -field generated by the finite dimensional sets, then enumerating the rationals $q_1, q_2, \dots$ and applying Kolmogorov's extension theorem shows that we can construct a probability ν_x on $(\Omega_q, \mathcal{F}_q)$ so that $\nu_x\{\omega : \omega(0) = x\} = 1$ and (7.1.3) holds when the $t_i \in \mathbf{Q}_2$. To extend B_t to a process defined on $[0, \infty)$, we will show:

Theorem 7.1.2 *Let $T < \infty$ and $x \in \mathbf{R}$. ν_x assigns probability one to paths $\omega : \mathbf{Q}_2 \rightarrow \mathbf{R}$ that are uniformly continuous on $\mathbf{Q}_2 \cap [0, T]$.*

Remark It will take quite a bit of work to prove Theorem 7.1.2. Before taking on that task, we will attend to the last measure theoretic detail: We tidy things up by moving our probability measures to $(C, \mathcal{C})$, where $C = \{\text{continuous } \omega : [0, \infty) \rightarrow \mathbf{R}\}$ and $\mathcal{C}$ is the σ -field generated by the coordinate maps $t \rightarrow \omega(t)$. To do this, we observe that the map ψ that takes a uniformly continuous point in Ω_q to its unique continuous extension in C is measurable, and we set

$$P_x = \nu_x \circ \psi^{-1}$$

Our construction guarantees that $B_t(\omega) = \omega_t$ has the right finite dimensional distributions for $t \in \mathbf{Q}_2$. Continuity of paths and a simple limiting argument show that this is true when $t \in [0, \infty)$. Finally, the reader should note that, as in the case of Markov chains, we have one set of random variables $B_t(\omega) = \omega(t)$, and a family of probability measures $P_x, x \in \mathbf{R}$, so that under P_x, B_t is a Brownian motion with $P_x(B_0 = x) = 1$.

Proof By translation invariance and scaling (7.1.1), we can without loss of generality suppose $B_0 = 0$ and prove the result for $T = 1$. In this case, part (b) of the definition and the scaling relation imply

$$E_0(|B_t - B_s|)^4 = E_0|B_{t-s}|^4 = C(t-s)^2$$

where $C = E_0|B_1|^4 < \infty$. From the last observation, we get the desired uniform continuity by using the following result due to Kolmogorov. Thanks to Robin Pemantle for simplifying the proof and to Timo Seppäläinen for correcting the simplification.

Theorem 7.1.3 Suppose $E|X_s - X_t|^\beta \leq K|t - s|^{1+\alpha}$, where $\alpha, \beta > 0$. If $\gamma < \alpha/\beta$, then with probability one there is a constant $C(\omega)$ so that

$$|X(q) - X(r)| \leq C|q - r|^\gamma \quad \text{for all } q, r \in \mathbf{Q}_2 \cap [0, 1]$$

Proof Let $G_n = \{|X(i/2^n) - X((i-1)/2^n)| \leq 2^{-\gamma n} \text{ for all } 0 < i \leq 2^n\}$. Chebyshev's inequality implies $P(|Y| > a) \leq a^{-\beta} E|Y|^\beta$, so if we let $\lambda = \alpha - \beta\gamma > 0$, then

$$P(G_n^c) \leq 2^n \cdot 2^{n\beta\gamma} \cdot E|X(j2^{-n}) - X(i2^{-n})|^\beta = K2^{-n\lambda}$$

Lemma 7.1.4 On $H_N = \cap_{n=N}^\infty G_n$ we have

$$|X(q) - X(r)| \leq \frac{3}{1 - 2^{-\gamma}} |q - r|^\gamma$$

for $q, r \in \mathbf{Q}_2 \cap [0, 1]$ with $|q - r| < 2^{-N}$.

Proof of Lemma 7.1.4 Let $q, r \in \mathbf{Q}_2 \cap [0, 1]$ with $0 < r - q < 2^{-N}$. Let $I_i^k = [(i-1)/2^k, i/2^k)$ and let m be the smallest value of k for which q and r are in different intervals I_i^k . Since they were in the same interval on level $k-1$, $q \in I_i^m$ and $r \in I_{i+1}^m$. We can write

$$\begin{aligned} r &= i2^{-m} + 2^{-r(1)} + \dots + 2^{-r(\ell)} \\ q &= i2^{-m} - 2^{-q(1)} - \dots - 2^{-q(k)} \end{aligned}$$

where $N < r(1) < \dots < r(\ell)$ and $N < q(1) < \dots < q(k)$. To see this, note that as we further subdivide, r will lie in the left or right half. When it is in the right half we add another term.

On H_N

$$|X(q) - X((i-1)2^{-m})| \leq \sum_{h=1}^k (2^{-q(h)})^\gamma \leq \sum_{h=m}^{\infty} (2^{-\gamma})^h = \frac{2^{-\gamma m}}{1 - 2^{-\gamma}}$$

$$|X(r) - X(i2^{-m})| \leq \frac{2^{-\gamma m}}{1 - 2^{-\gamma}}$$

Combining the last three inequalities with $2^{-m} \leq |q - r|$ and $1 - 2^{-\gamma} > 1$ completes the proof of Lemma 7.1.4. $\square$

To prove Theorem 7.1.3 now, we note that

$$P(H_N^c) \leq \sum_{n=N}^{\infty} P(G_n^c) \leq K \sum_{n=N}^{\infty} 2^{-n\lambda} = K 2^{-N\lambda} / (1 - 2^{-\lambda})$$

Since $\sum_{N=1}^{\infty} P(H_N^c) < \infty$, the Borel-Cantelli lemma, Theorem 2.3.1, implies

$$|X(q) - X(r)| \leq A|q - r|^\gamma \quad \text{for } q, r \in \mathbf{Q}_2 \text{ with } |q - r| < \delta(\omega).$$

To extend this to $q, r \in \mathbf{Q}_2 \cap [0, 1]$, let $s_0 = q < s_1 < \dots < s_n = r$ with $|s_i - s_{i-1}| < \delta(\omega)$ and use the triangle inequality to conclude $|X(q) - X(r)| \leq C(\omega)|q - r|^\gamma$, where $C(\omega) = 1 + \delta(\omega)^{-1}$. $\square$

The scaling relation, (7.1.1), implies

$$E|B_t - B_s|^{2m} = C_m |t - s|^m \quad \text{where } C_m = E|B_1|^{2m}$$

so using Theorem 7.1.3 with $\beta = 2m$, $\alpha = m - 1$ and letting $m \rightarrow \infty$ gives a result of Wiener (1923).

Theorem 7.1.5 *Brownian paths are Hölder continuous for any exponent $\gamma < 1/2$.*

It is easy to show:

Theorem 7.1.6 *With probability one, Brownian paths are not Lipschitz continuous (and hence not differentiable) at any point.*

Remark The nondifferentiability of Brownian paths was discovered by Paley, Wiener, and Zygmund (1933). Paley died in 1933 at the age of 26 in a skiing accident while the paper was in press. The proof we are about to give is due to Dvoretzky, Erdős, and Kakutani (1961).

Proof Fix a constant $C < \infty$ and let $A_n = \{\omega : \text{there is an } s \in [0, 1] \text{ so that } |B_t - B_s| \leq C|t - s| \text{ when } |t - s| \leq 3/n\}$. For $1 \leq k \leq n - 2$, let

$$Y_{k,n} = \max \left\{ \left| B\left(\frac{k+j}{n}\right) - B\left(\frac{k+j-1}{n}\right) \right| : j = 0, 1, 2 \right\}$$

$$B_n = \{ \text{at least one } Y_{k,n} \leq 5C/n \}$$

The triangle inequality implies $A_n \subset B_n$. The worst case is $s = 1$. We pick $k = n - 2$ and observe

$$\begin{aligned} \left| B\left(\frac{n-3}{n}\right) - B\left(\frac{n-2}{n}\right) \right| &\leq \left| B\left(\frac{n-3}{n}\right) - B(1) \right| + \left| B(1) - B\left(\frac{n-2}{n}\right) \right| \\ &\leq C(3/n + 2/n) \end{aligned}$$

Using $A_n \subset B_n$ and the scaling relation (7.1.1) now gives

$$\begin{aligned} P(A_n) &\leq P(B_n) \leq nP(|B(1/n)| \leq 5C/n)^3 = nP(|B(1)| \leq 5C/n^{1/2})^3 \\ &\leq n\{(10C/n^{1/2}) \cdot (2\pi)^{-1/2}\}^3 \end{aligned}$$

since $\exp(-x^2/2) \leq 1$. Letting $n \rightarrow \infty$ shows $P(A_n) \rightarrow 0$. Noticing $n \rightarrow A_n$ is increasing shows $P(A_n) = 0$ for all n and completes the proof. $\square$

Remark Let $\mathcal{H}_\gamma(\omega)$ be the set of times at which the path $\omega \in C$ is Hölder continuous of order γ . Theorem 7.1.5 shows that $P(\mathcal{H}_\gamma = [0, \infty)) = 1$ for $\gamma < 1/2$. Exercise 7.1.5 shows that $P(\mathcal{H}_\gamma = \emptyset) = 1$ for $\gamma > 1/2$. Exercise 7.2.4 shows $P(t \in \mathcal{H}_{1/2}) = 0$ for each t , but B. Davis (1983) has shown $P(\mathcal{H}_{1/2} \neq \emptyset) = 1$. Perkins (1983) has computed the Hausdorff dimension of

$$\left\{ t \in (0, 1) : \limsup_{h \downarrow 0} \frac{|B_{t+h} - B_t|}{h^{1/2}} \leq c \right\}$$

Multidimensional Brownian Motion

All of the results in this section have been for one-dimensional Brownian motion. To define a d -dimensional Brownian motion starting at $x \in \mathbf{R}^d$ we let $B_t^1, \dots, B_t^d$ be independent Brownian motions with $B_0^i = x_i$. As in the case $d = 1$, these are realized as probability measures P_x on $(C, \mathcal{C})$ where $C = \{\text{continuous } \omega : [0, \infty) \rightarrow \mathbf{R}^d\}$ and $\mathcal{C}$ is the σ -field generated by the coordinate maps. Since the coordinates are independent, it is easy to see that the finite dimensional distributions satisfy (7.1.2) with transition probability

$$p_t(x, y) = (2\pi t)^{-d/2} \exp(-|y - x|^2/2t) \quad (7.1.4)$$

Exercises

- 7.1.1 Given $s < t$ find $P(B(s) > 0, B(t) > 0)$. Hint: write the probability in terms of $B(s)$ and $B(t) - B(s)$.
- 7.1.2 Find $E(B_1^2 B_2 B_3)$.
- 7.1.3 Let $W = \int_0^t B_s ds$. Find EW and EW^2 . What is the distribution of W ?
- 7.1.4 $A \in \mathcal{F}_o$ if and only if there is a sequence of times $t_1, t_2, \dots$ in $[0, \infty)$ and a $B \in \mathcal{R}^{\{1, 2, \dots\}}$ so that $A = \{\omega : (\omega(t_1), \omega(t_2), \dots) \in B\}$. In words, all events in $\mathcal{F}_o$ depend on only countably many coordinates.
- 7.1.5 Looking at the proof of Theorem 7.1.6 carefully shows that if $\gamma > 5/6$, then B_t is not Hölder continuous with exponent γ at any point in $[0, 1]$. Show, by considering k increments instead of 3, that the last conclusion is true for all $\gamma > 1/2 + 1/k$.

7.1.6 Fix t and let $\Delta_{m,n} = B(tm2^{-n}) - B(t(m-1)2^{-n})$. Compute

$$E\left(\sum_{m \leq 2^n} \Delta_{m,n}^2 - t\right)^2$$

and use Borel-Cantelli to conclude that $\sum_{m \leq 2^n} \Delta_{m,n}^2 \rightarrow t$ a.s. as $n \rightarrow \infty$.

Remark The last result is true if we consider a sequence of partitions $\Pi_1 \subset \Pi_2 \subset \dots$ with mesh $\rightarrow 0$. See Freedman (1971a) pp. 42–46. However, the true quadratic variation, defined as the sup over all partitions, is ∞ .

7.2 Markov Property, Blumenthal's 0-1 Law

Intuitively, the Markov property says “if $s \geq 0$ then $B(t+s) - B(s)$, $t \geq 0$ is a Brownian motion that is independent of what happened before time s .” The first step in making this into a precise statement is to explain what we mean by “what happened before time s .” The first thing that comes to mind is

$$\mathcal{F}_s^o = \sigma(B_r : r \leq s)$$

For reasons that will become clear as we go along, it is convenient to replace $\mathcal{F}_s^o$ by

$$\mathcal{F}_s^+ = \cap_{t>s} \mathcal{F}_t^o$$

The fields $\mathcal{F}_s^+$ are nicer because they are **right continuous**:

$$\cap_{t>s} \mathcal{F}_t^+ = \cap_{t>s} (\cap_{u>t} \mathcal{F}_u^o) = \cap_{u>s} \mathcal{F}_u^o = \mathcal{F}_s^+$$

In words, the $\mathcal{F}_s^+$ allow us an “infinitesimal peek at the future,” i.e., $A \in \mathcal{F}_s^+$ if it is in $\mathcal{F}_{s+\epsilon}^o$ for any $\epsilon > 0$. If $f(u) > 0$ for all $u > 0$, then in $d = 1$ the random variable

$$\limsup_{t \downarrow s} \frac{B_t - B_s}{f(t-s)}$$

is measurable with respect to $\mathcal{F}_s^+$ but not $\mathcal{F}_s^o$. We will see that there are no interesting examples, i.e., $\mathcal{F}_s^+$ and $\mathcal{F}_s^o$ are the same (up to null sets).

To state the Markov property, we need some notation. Recall that we have a family of measures P_x , $x \in \mathbf{R}^d$, on $(C, \mathcal{C})$ so that under P_x , $B_t(\omega) = \omega(t)$ is a Brownian motion starting at x . For $s \geq 0$, we define the **shift transformation** $\theta_s : C \rightarrow C$ by

$$(\theta_s \omega)(t) = \omega(s+t) \quad \text{for } t \geq 0$$

In words, we cut off the part of the path before time s and then shift the path so that time s becomes time 0.

Theorem 7.2.1 (Markov property) *If $s \geq 0$ and Y is bounded and $\mathcal{C}$ measurable, then for all $x \in \mathbf{R}^d$*

$$E_x(Y \circ \theta_s | \mathcal{F}_s^+) = E_{B_s} Y$$

where the right-hand side is the function $\varphi(x) = E_x Y$ evaluated at $x = B_s$.

Proof By the definition of conditional expectation, what we need to show is that

$$E_x(Y \circ \theta_s; A) = E_x(E_{B_s} Y; A) \quad \text{for all } A \in \mathcal{F}_s^+ \quad (7.2.1)$$

We will begin by proving the result for a carefully chosen special case and then use the monotone class theorem (MCT) to get the general case. Suppose $Y(\omega) = \prod_{1 \leq m \leq n} f_m(\omega(t_m))$, where $0 < t_1 < \dots < t_n$ and the f_m are bounded and measurable. Let $0 < h < t_1$, let $0 < s_1 \dots < s_k \leq s + h$, and let $A = \{\omega : \omega(s_j) \in A_j, 1 \leq j \leq k\}$, where $A_j \in \mathcal{R}$ for $1 \leq j \leq k$. From the definition of Brownian motion, it follows that

$$\begin{aligned} E_x(Y \circ \theta_s; A) &= \int_{A_1} dx_1 p_{s_1}(x, x_1) \int_{A_2} dx_2 p_{s_2-s_1}(x_1, x_2) \cdots \\ &\quad \int_{A_k} dx_k p_{s_k-s_{k-1}}(x_{k-1}, x_k) \int dy p_{s+h-s_k}(x_k, y) \varphi(y, h) \end{aligned}$$

where

$$\varphi(y, h) = \int dy_1 p_{t_1-h}(y, y_1) f_1(y_1) \cdots \int dy_n p_{t_n-t_{n-1}}(y_{n-1}, y_n) f_n(y_n)$$

For more details, see the proof of (5.2.3), which applies without change here. Using that identity on the right-hand side, we have

$$E_x(Y \circ \theta_s; A) = E_x(\varphi(B_{s+h}, h); A) \quad (7.2.2)$$

The last equality holds for all finite dimensional sets A , so the $\pi - \lambda$ theorem, Theorem 2.1.6, implies that it is valid for all $A \in \mathcal{F}_{s+h}^o \supset \mathcal{F}_s^+$.

It is easy to see by induction on n that

$$\begin{aligned} \psi(y_1) &= f_1(y_1) \int dy_2 p_{t_2-t_1}(y_1, y_2) f_2(y_2) \\ &\quad \cdots \int dy_n p_{t_n-t_{n-1}}(y_{n-1}, y_n) f_n(y_n) \end{aligned}$$

is bounded and measurable. Letting $h \downarrow 0$ and using the dominated convergence theorem shows that if $x_h \rightarrow x$, then

$$\phi(x_h, h) = \int dy_1 p_{t_1-h}(x_h, y_1) \psi(y_1) \rightarrow \phi(x, 0)$$

as $h \downarrow 0$. Using (7.2.2) and the bounded convergence theorem now gives

$$E_x(Y \circ \theta_s; A) = E_x(\varphi(B_s, 0); A)$$

for all $A \in \mathcal{F}_s^+$. This shows that (7.2.1) holds for $Y = \prod_{1 \leq m \leq n} f_m(\omega(t_m))$ and the f_m are bounded and measurable.

The desired conclusion now follows from the monotone class theorem, 5.2.2. Let $\mathcal{H}$ = the collection of bounded functions for which (7.2.1) holds. $\mathcal{H}$ clearly has properties (ii) and (iii). Let $\mathcal{A}$ be the collection of sets of the form $\{\omega : \omega(t_j) \in A_j\}$, where $A_j \in \mathcal{R}$. The special case treated here shows (i) holds and the desired conclusion follows. $\square$

The reader will see many applications of the Markov property later, so we turn our attention now to a “triviality” that has surprising consequences. Since

$$E_x(Y \circ \theta_s | \mathcal{F}_s^+) = E_{B(s)} Y \in \mathcal{F}_s^o$$

it follows from Theorem 4.1.12 that

$$E_x(Y \circ \theta_s | \mathcal{F}_s^+) = E_x(Y \circ \theta_s | \mathcal{F}_s^o)$$

From the last equation, it is a short step to:

Theorem 7.2.2 *If $Z \in \mathcal{C}$ is bounded, then for all $s \geq 0$ and $x \in \mathbf{R}^d$,*

$$E_x(Z | \mathcal{F}_s^+) = E_x(Z | \mathcal{F}_s^o)$$

Proof As in the proof of Theorem 7.2.1, it suffices to prove the result when

$$Z = \prod_{m=1}^n f_m(B(t_m))$$

and the f_m are bounded and measurable. In this case, Z can be written as $X(Y \circ \theta_s)$, where $X \in \mathcal{F}_s^o$ and Y is $\mathcal{C}$ measurable, so

$$E_x(Z | \mathcal{F}_s^+) = X E_x(Y \circ \theta_s | \mathcal{F}_s^+) = X E_{B_s} Y \in \mathcal{F}_s^o$$

and the proof is complete. $\square$

If we let $Z \in \mathcal{F}_s^+$, then Theorem 7.2.2 implies $Z = E_x(Z | \mathcal{F}_s^o) \in \mathcal{F}_s^o$, so the two σ -fields are the same up to null sets. At first glance, this conclusion is not exciting. The fun starts when we take $s = 0$ in Theorem 7.2.2 to get:

Theorem 7.2.3 (Blumenthal's 0-1 law) *If $A \in \mathcal{F}_0^+$, then for all $x \in \mathbf{R}^d$,*

$$P_x(A) \in \{0, 1\}.$$

Proof Using $A \in \mathcal{F}_0^+$, Theorem 7.2.2, and $\mathcal{F}_0^o = \sigma(B_0)$ is trivial under P_x gives

$$1_A = E_x(1_A | \mathcal{F}_0^+) = E_x(1_A | \mathcal{F}_0^o) = P_x(A) \quad P_x \text{ a.s.}$$

This shows that the indicator function 1_A is a.s. equal to the number $P_x(A)$, and the result follows. $\square$

In words, the last result says that the **germ field**, $\mathcal{F}_0^+$, is trivial. This result is very useful in studying the local behavior of Brownian paths. For the rest of the section we restrict our attention to $d = 1$.

Theorem 7.2.4 *If $\tau = \inf\{t \geq 0 : B_t > 0\}$, then $P_0(\tau = 0) = 1$.*

Proof $P_0(\tau \leq t) \geq P_0(B_t > 0) = 1/2$, since the normal distribution is symmetric about 0. Letting $t \downarrow 0$, we conclude

$$P_0(\tau = 0) = \lim_{t \downarrow 0} P_0(\tau \leq t) \geq 1/2$$

so it follows from Theorem 7.2.3 that $P_0(\tau = 0) = 1$. $\square$

Once Brownian motion must hit $(0, \infty)$ immediately starting from 0, it must also hit $(-\infty, 0)$ immediately. Since $t \rightarrow B_t$ is continuous, this forces:

Theorem 7.2.5 *If $T_0 = \inf\{t > 0 : B_t = 0\}$ then $P_0(T_0 = 0) = 1$.*

Theorem 7.2.3 concerns the behavior of B_t as $t \rightarrow 0$. By using a trick, we can use this result to get information about the behavior as $t \rightarrow \infty$.

Theorem 7.2.6 *If B_t is a Brownian motion starting at 0, then so is the process defined by $X_0 = 0$ and $X_t = tB(1/t)$ for $t > 0$.*

Proof Here we will check the second definition of Brownian motion. To do this, we note:
(i) If $0 < t_1 < \dots < t_n$, then $(X(t_1), \dots, X(t_n))$ has a multivariate normal distribution with mean 0. (ii) $EX_s = 0$ and if $s < t$, then

$$E(X_s X_t) = stE(B(1/s)B(1/t)) = s$$

For (iii) we note that X is clearly continuous at $t \neq 0$.

To handle $t = 0$, we begin by observing that the strong law of large numbers implies $B_n/n \rightarrow 0$ as $n \rightarrow \infty$ through the integers. To handle values in between integers, we note that Kolmogorov's inequality, Theorem 2.5.5, implies

$$P\left(\sup_{0 < k \leq 2^m} |B(n + k2^{-m}) - B_n| > n^{2/3}\right) \leq n^{-4/3} E(B_{n+1} - B_n)^2$$

Letting $m \rightarrow \infty$, we have

$$P\left(\sup_{u \in [n, n+1]} |B_u - B_n| > n^{2/3}\right) \leq n^{-4/3}$$

Since $\sum_n n^{-4/3} < \infty$, the Borel-Cantelli lemma implies $B_u/u \rightarrow 0$ as $u \rightarrow \infty$. Taking $u = 1/t$, we have $X_t \rightarrow 0$ as $t \rightarrow 0$. $\square$

Theorem 7.2.6 allows us to relate the behavior of B_t as $t \rightarrow \infty$ and as $t \rightarrow 0$. Combining this idea with Blumenthal's 0-1 law leads to a very useful result. Let

$$\begin{aligned}\mathcal{F}'_t &= \sigma(B_s : s \geq t) = \text{the future at time } t \\ \mathcal{T} &= \cap_{t \geq 0} \mathcal{F}'_t = \text{the tail } \sigma\text{-field.}\end{aligned}$$

Theorem 7.2.7 *If $A \in \mathcal{T}$, then either $P_x(A) \equiv 0$ or $P_x(A) \equiv 1$.*

Remark Notice that this is stronger than the conclusion of Blumenthal's 0-1 law. The examples $A = \{\omega : \omega(0) \in D\}$ show that for A in the germ σ -field $\mathcal{F}_0^+$, the value of $P_x(A)$, $1_D(x)$ in this case, may depend on x .

Proof Since the tail σ -field of B is the same as the germ σ -field for X , it follows that $P_0(A) \in \{0, 1\}$. To improve this to the conclusion given, observe that $A \in \mathcal{F}'_1$, so 1_A can be written as $1_D \circ \theta_1$. Applying the Markov property gives

$$\begin{aligned}P_x(A) &= E_x(1_D \circ \theta_1) = E_x(E_x(1_D \circ \theta_1 | \mathcal{F}_1)) = E_x(E_{B_1} 1_D) \\ &= \int (2\pi)^{-1/2} \exp(-(y-x)^2/2) P_y(D) dy\end{aligned}$$

Taking $x = 0$, we see that if $P_0(A) = 0$, then $P_y(D) = 0$ for a.e. y with respect to Lebesgue measure, and using the formula again shows $P_x(A) = 0$ for all x . To handle the case $P_0(A) = 1$, observe that $A^c \in \mathcal{T}$ and $P_0(A^c) = 0$, so the last result implies $P_x(A^c) = 0$ for all x . $\square$

The next result is a typical application of Theorem 7.2.7.

Theorem 7.2.8 *Let B_t be a one-dimensional Brownian motion starting at 0 then with probability 1,*

$$\limsup_{t \rightarrow \infty} B_t / \sqrt{t} = \infty \quad \liminf_{t \rightarrow \infty} B_t / \sqrt{t} = -\infty$$

Proof Let $K < \infty$. By Exercise 2.3.1 and scaling

$$P_0(B_n / \sqrt{n} \geq K \text{ i.o.}) \geq \limsup_{n \rightarrow \infty} P_0(B_n \geq K\sqrt{n}) = P_0(B_1 \geq K) > 0$$

so the 0–1 law in Theorem 7.2.7 implies the probability is 1. Since K is arbitrary, this proves the first result. The second one follows from symmetry. $\square$

From Theorem 7.2.8, translation invariance, and the continuity of Brownian paths it follows that we have:

Theorem 7.2.9 *Let B_t be a one-dimensional Brownian motion and let $A = \cap_n \{B_t = 0 \text{ for some } t \geq n\}$. Then $P_x(A) = 1$ for all x .*

In words, one-dimensional Brownian motion is recurrent. For any starting point x , it will return to 0 “infinitely often,” i.e., there is a sequence of times $t_n \uparrow \infty$ so that $B_{t_n} = 0$. We have to be careful with the interpretation of the phrase in quotes, since starting from 0, B_t will hit 0 infinitely many times by time $\epsilon > 0$.

Last rites With our discussion of Blumenthal's 0-1 law complete, the distinction between $\mathcal{F}_s^+$ and $\mathcal{F}_s^o$ is no longer important, so we will make one final improvement in our σ -fields and remove the superscripts. Let

$$\begin{aligned} \mathcal{N}_x &= \{A : A \subset D \text{ with } P_x(D) = 0\} \\ \mathcal{F}_s^x &= \sigma(\mathcal{F}_s^+ \cup \mathcal{N}_x) \\ \mathcal{F}_s &= \cap_x \mathcal{F}_s^x \end{aligned}$$

$\mathcal{N}_x$ are the **null sets** and $\mathcal{F}_s^x$ are the completed σ -fields for P_x . Since we do not want the filtration to depend on the initial state, we take the intersection of all the σ -fields. The reader should note that it follows from the definition that the $\mathcal{F}_s$ are right continuous.

Exercises

7.2.1 Let $T_0 = \inf\{s > 0 : B_s = 0\}$ and let $R = \inf\{t > 1 : B_t = 0\}$. R is for right or return. Use the Markov property at time 1 to get

$$P_x(R > 1 + t) = \int p_1(x, y) P_y(T_0 > t) dy \quad (7.2.3)$$

7.2.2 Let $T_0 = \inf\{s > 0 : B_s = 0\}$ and let $L = \sup\{t \leq 1 : B_t = 0\}$. L is for left or last. Use the Markov property at time $0 < t < 1$ to conclude

$$P_0(L \leq t) = \int p_t(0, y) P_y(T_0 > 1 - t) dy \quad (7.2.4)$$

7.2.3 If $a < b$, then with probability one a is the limit of local maximum of B_t in (a, b) . So the set of local maxima of B_t is almost surely a dense set. However, unlike the zero set it is countable.

7.2.4 (i) Suppose $f(t) > 0$ for all $t > 0$. Use Theorem 7.2.3 to conclude that $\limsup_{t \downarrow 0} B(t)/f(t) = c$, P_0 a.s., where $c \in [0, \infty]$ is a constant. (ii) Show that if $f(t) = \sqrt{t}$, then $c = \infty$, so with probability one Brownian paths are not Hölder continuous of order $1/2$ at 0.

7.3 Stopping Times, Strong Markov Property

Generalizing the definition from discrete time, we call a random variable S taking values in $[0, \infty]$ a **stopping time** if for all $t \geq 0$, $\{S < t\} \in \mathcal{F}_t$. In the last definition, we have obviously made a choice between $\{S < t\}$ and $\{S \leq t\}$. This makes a big difference in discrete time but none in continuous time (for a right continuous filtration $\mathcal{F}_t$) :

If $\{S \leq t\} \in \mathcal{F}_t$ then $\{S < t\} = \cup_n \{S \leq t - 1/n\} \in \mathcal{F}_t$.

If $\{S < t\} \in \mathcal{F}_t$ then $\{S \leq t\} = \cap_n \{S < t + 1/n\} \in \mathcal{F}_t$.

The first conclusion requires only that $t \rightarrow \mathcal{F}_t$ is increasing. The second relies on the fact that $t \rightarrow \mathcal{F}_t$ is right continuous. Theorem 7.3.2 and 7.3.3 show that when checking something is a stopping time, it is nice to know that the two definitions are equivalent.

Theorem 7.3.1 *If G is an open set and $T = \inf\{t \geq 0 : B_t \in G\}$, then T is a stopping time.*

Proof Since G is open and $t \rightarrow B_t$ is continuous, $\{T < t\} = \cup_{q < t} \{B_q \in G\}$, where the union is over all rational q , so $\{T < t\} \in \mathcal{F}_t$. Here we need to use the rationals to get a countable union, and hence a measurable set. $\square$

Theorem 7.3.2 *If T_n is a sequence of stopping times and $T_n \downarrow T$, then T is a stopping time.*

Proof $\{T < t\} = \cup_n \{T_n < t\}$. $\square$

Theorem 7.3.3 *If T_n is a sequence of stopping times and $T_n \uparrow T$, then T is a stopping time.*

Proof $\{T \leq t\} = \cap_n \{T_n \leq t\}$. $\square$

Theorem 7.3.4 *If K is a closed set and $T = \inf\{t \geq 0 : B_t \in K\}$, then T is a stopping time.*

Proof Let $B(x, r) = \{y : |y - x| < r\}$, let $G_n = \cup_{x \in K} B(x, 1/n)$, and let $T_n = \inf\{t \geq 0 : B_t \in G_n\}$. Since G_n is open, it follows from Theorem 7.3.1 that T_n is a stopping time. I claim that as $n \uparrow \infty$, $T_n \uparrow T$. To prove this, notice that $T \geq T_n$ for all n , so $\lim T_n \leq T$. To prove $T \leq \lim T_n$, we can suppose that $T_n \uparrow t < \infty$. Since $B(T_n) \in \bar{G}_n$ for all n and $B(T_n) \rightarrow B(t)$, it follows that $B(t) \in K$ and $T \leq t$. $\square$

Theorems 7.3.1 and 7.3.4 will take care of all the hitting times we will consider. Using 7.3.2 with the second result, one can show that the hitting time of a union of closed sets is a stopping time. It is known that the hitting time of a Borel set is a stopping time but proving that is very difficult.

We will need the next result in the proof of the strong Markov property.

Theorem 7.3.5 *Let S be a stopping time and let $S_n = ([2^n S] + 1)/2^n$, where $[x]$ = the largest integer $\leq x$. That is,*

$$S_n = (m + 1)2^{-n} \text{ if } m2^{-n} \leq S < (m + 1)2^{-n}$$

In words, we stop at the first time of the form $k2^{-n}$ after S (i.e., $> S$). From the verbal description, it should be clear that S_n is a stopping time.

Proof If $m/2^{-n} < \leq (m + 1)/2^{-n}$, then $\{S_n < t\} = \{S \leq m/2^n\} \in \mathcal{F}_{m/2^n} \subset \mathcal{F}_t$. It is impossible to have $S_n = 0$. $\square$

Our next goal is to state and prove the strong Markov property. To do this, we need to generalize two definitions from Section 4.1. Given a nonnegative random variable $S(\omega)$ we define the random shift θ_S , which “cuts off the part of ω before $S(\omega)$ and then shifts the path so that time $S(\omega)$ becomes time 0.” In symbols, we set

$$(\theta_S \omega)(t) = \begin{cases} \omega(S(\omega) + t) & \text{on } \{S < \infty\} \\ \Delta & \text{on } \{S = \infty\} \end{cases}$$

where Δ is an extra point we add to C . As in Section 6.3, we will usually explicitly restrict our attention to $\{S < \infty\}$, so the reader does not have to worry about the second half of the definition.

The second quantity $\mathcal{F}_S$, “the information known at time S ,” is a little more subtle. Imitating the discrete time definition from Section 4.1, we let

$$\mathcal{F}_S = \{A : A \cap \{S \leq t\} \in \mathcal{F}_t \text{ for all } t \geq 0\}$$

In words, this makes the reasonable demand that the part of A that lies in $\{S \leq t\}$ should be measurable with respect to the information available at time t . Again we have made a choice between $\leq t$ and $< t$, but as in the case of stopping times, this makes no difference, and it is useful to know that the two definitions are equivalent.

Remark When $\mathcal{F}_t$ is right continuous, the last definition is unchanged if we replace $\{S \leq t\}$ by $\{S < t\}$.

We will now derive some properties of $\mathcal{F}_S$. The first is intuitively obvious: at a later time we have more information.

Theorem 7.3.6 *If $S \leq T$ are stopping times, then $\mathcal{F}_S \subset \mathcal{F}_T$.*

Proof If $A \in \mathcal{F}_S$, then $A \cap \{T \leq t\} = (A \cap \{S \leq t\}) \cap \{T \leq t\} \in \mathcal{F}_t$. $\square$

Theorem 7.3.7 *If $T_n \downarrow T$ are stopping times, then $\mathcal{F}_T = \cap \mathcal{F}(T_n)$.*

Proof Theorem 7.3.6 implies $\mathcal{F}(T_n) \supset \mathcal{F}_T$ for all n . To prove the other inclusion, let $A \in \cap \mathcal{F}(T_n)$. Since $A \cap \{T_n < t\} \in \mathcal{F}_t$ and $T_n \downarrow T$, it follows that $A \cap \{T < t\} \in \mathcal{F}_t$. $\square$

The next result is obvious but takes a little work.

Theorem 7.3.8 $B_S \in \mathcal{F}_S$, i.e., the value of B_S is measurable with respect to the information known at time S !

Proof To prove this, let $S_n = ([2^n S] + 1)/2^n$ be the stopping times defined in Theorem 7.3.5. If A is a Borel set.

$$\{B(S_n) \in A\} = \cup_{m=1}^{\infty} \{S_n = m/2^n, B(m/2^n) \in A\} \in \mathcal{F}_{S_n}$$

Now let $n \rightarrow \infty$ and use Theorem 7.3.7. □

We are now ready to state the strong Markov property, which says that the Markov property holds at stopping times. It is interesting that the notion of Brownian motion dates to the very beginning of the twentieth century, but the first proofs of the strong Markov property were given independently by Hunt (1956), and Dynkin and Yushkevich (1956). Hunt writes “Although mathematicians use this extended Markoff property, at least as a heuristic principle, I have nowhere found it discussed with rigor.”

Theorem 7.3.9 (Strong Markov property) Let $(s, \omega) \rightarrow Y_s(\omega)$ be bounded and $\mathcal{R} \times \mathcal{C}$ measurable. If S is a stopping time, then for all $x \in \mathbb{R}^d$

$$E_x(Y_S \circ \theta_S | \mathcal{F}_S) = E_{B(S)} Y_S \text{ on } \{S < \infty\}$$

where the right-hand side is the function $\varphi(x, t) = E_x Y_t$ evaluated at $x = B(S)$, $t = S$.

Remark The only facts about Brownian motion used here are that (i) it is a Markov process, and (ii) if f is bounded and continuous, then $x \rightarrow E_x f(B_t)$ is continuous. In Markov process theory (ii) is called the Feller property. Hunt’s proof only applies to Brownian motion, and Dynkin and Yushkevich proved the result in this generality.

Proof We first prove the result under the assumption that there is a sequence of times $t_n \uparrow \infty$, so that $P_x(S < \infty) = \sum P_x(S = t_n)$. In this case, the proof is basically the same as the proof of Theorem 5.2.5. We break things down according to the value of S , apply the Markov property, and put the pieces back together. If we let $Z_n = Y_{t_n}(\omega)$ and $A \in \mathcal{F}_S$, then

$$E_x(Y_S \circ \theta_S; A \cap \{S < \infty\}) = \sum_{n=1}^{\infty} E_x(Z_n \circ \theta_{t_n}; A \cap \{S = t_n\})$$

Now if $A \in \mathcal{F}_S$, $A \cap \{S = t_n\} = (A \cap \{S \leq t_n\}) - (A \cap \{S \leq t_{n-1}\}) \in \mathcal{F}_{t_n}$, so it follows from the Markov property that the sum is

$$= \sum_{n=1}^{\infty} E_x(E_{B(t_n)} Z_n; A \cap \{S = t_n\}) = E_x(E_{B(S)} Y_S; A \cap \{S < \infty\})$$

To prove the result in general, we let $S_n = ([2^n S] + 1)/2^n$ be the stopping time defined in Exercise 7.3.5. To be able to let $n \rightarrow \infty$, we restrict our attention to Y ’s of the form

$$Y_s(\omega) = f_0(s) \prod_{m=1}^n f_m(\omega(t_m)) \quad (7.3.1)$$

where $0 < t_1 < \dots < t_n$ and $f_0, \dots, f_n$ are bounded and continuous. If f is bounded and continuous, then the dominated convergence theorem implies that

$$x \rightarrow \int dy p_t(x, y) f(y)$$

is continuous. From this and induction, it follows that

$$\begin{aligned} \varphi(x, s) &= E_x Y_s = f_0(s) \int dy_1 p_{t_1}(x, y_1) f_1(y_1) \\ &\quad \dots \int dy_n p_{t_n - t_{n-1}}(y_{n-1}, y_n) f_n(y_n) \end{aligned}$$

is bounded and continuous.

Having assembled the necessary ingredients, we can now complete the proof. Let $A \in \mathcal{F}_S$. Since $S \leq S_n$, Theorem 7.3.6 implies $A \in \mathcal{F}(S_n)$. Applying the special case proved here to S_n and observing that $\{S_n < \infty\} = \{S < \infty\}$ gives

$$E_x(Y_{S_n} \circ \theta_{S_n}; A \cap \{S < \infty\}) = E_x(\varphi(B(S_n), S_n); A \cap \{S < \infty\})$$

Now, as $n \rightarrow \infty$, $S_n \downarrow S$, $B(S_n) \rightarrow B(S)$, $\varphi(B(S_n), S_n) \rightarrow \varphi(B(S), S)$ and

$$Y_{S_n} \circ \theta_{S_n} \rightarrow Y_S \circ \theta_S$$

so the bounded convergence theorem implies that the result holds when Y has the form given in (7.3.1).

To complete the proof now, we will apply the monotone class theorem. As in the proof of Theorem 7.2.1, we let $\mathcal{H}$ be the collection of Y for which

$$E_x(Y_S \circ \theta_S; A) = E_x(E_{B(S)} Y_S; A) \quad \text{for all } A \in \mathcal{F}_S$$

and it is easy to see that (ii) and (iii) hold. This time, however, we take $\mathcal{A}$ to be the sets of the form $A = G_0 \times \{\omega : \omega(s_j) \in G_j, 1 \leq j \leq k\}$, where the G_j are open sets. To verify (i), we note that if $K_j = G_j^c$ and $f_j^n(x) = 1 \wedge n\rho(x, K_j)$, where $\rho(x, K) = \inf\{|x - y| : y \in K\}$, then f_j^n are continuous functions with $f_j^n \uparrow 1_{G_j}$ as $n \uparrow \infty$. The facts that

$$Y_s^n(\omega) = f_0^n(s) \prod_{j=1}^k f_j^n(\omega(s_j)) \in \mathcal{H}$$

and (iii) holds for $\mathcal{H}$ imply that $1_A \in \mathcal{H}$. This verifies (i) in the monotone class theorem and completes the proof. $\square$

Exercises

- 7.3.1 Let A be an F_σ , that is, a countable union of closed sets. Show that $T_A = \inf\{t : B_t \in A\}$ is a stopping time.
- 7.3.2 If S and T are stopping times, then $S \wedge T = \min\{S, T\}$, $S \vee T = \max\{S, T\}$, and $S + T$ are also stopping times. In particular, if $t \geq 0$, then $S \wedge t$, $S \vee t$, and $S + t$ are stopping times.

7.3.3 Let T_n be a sequence of stopping times. Show that

$$\sup_n T_n, \quad \inf_n T_n, \quad \limsup_n T_n, \quad \liminf_n T_n$$

are stopping times.

7.3.4 Let S be a stopping time, let $A \in \mathcal{F}_S$, and let $R = S$ on A and $R = \infty$ on A^c . Show that R is a stopping time.

7.3.5 Let S and T be stopping times. $\{S < T\}$, $\{S > T\}$, and $\{S = T\}$ are in $\mathcal{F}_S$ (and in $\mathcal{F}_T$).

7.3.6 Let S and T be stopping times. $\mathcal{F}_S \cap \mathcal{F}_T = \mathcal{F}_{S \wedge T}$.

7.4 Path Properties

In this section, we will use the strong Markov property to derive properties of the zero set $\{t : B_t = 0\}$, the hitting times $T_a = \inf\{t : B_t = a\}$, and $\max_{0 \leq s \leq t} B_s$ for one-dimensional Brownian motion.

7.4.1 Zeros of Brownian Motion

Let $R_t = \inf\{u > t : B_u = 0\}$ and let $T_0 = \inf\{u > 0 : B_u = 0\}$. Now Theorem 7.2.9 implies $P_x(R_t < \infty) = 1$, so $B(R_t) = 0$ and the strong Markov property and Theorem 7.2.5 imply

$$P_x(T_0 \circ \theta_{R_t} > 0 | \mathcal{F}_{R_t}) = P_0(T_0 > 0) = 0$$

Taking expected value of the last equation, we see that

$$P_x(T_0 \circ \theta_{R_t} > 0 \text{ for some rational } t) = 0$$

From this, it follows that if a point $u \in \mathcal{Z}(\omega) \equiv \{t : B_t(\omega) = 0\}$ is isolated on the left (i.e., there is a rational $t < u$ so that $(t, u) \cap \mathcal{Z}(\omega) = \emptyset$), then it is, with probability one,

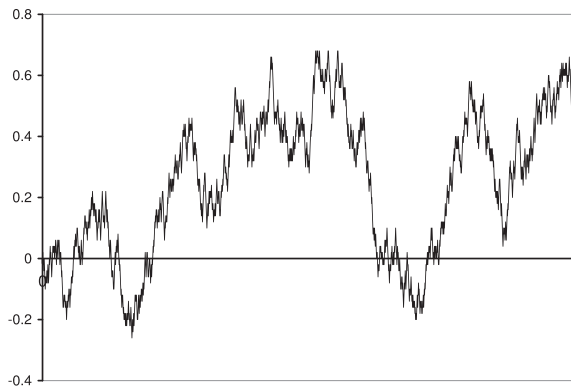


Figure 7.2 Simulation one-dimensional Brownian motion.

a decreasing limit of points in $\mathcal{Z}(\omega)$. This shows that the closed set $\mathcal{Z}(\omega)$ has no isolated points and hence must be uncountable. For the last step, see Hewitt and Stromberg (1965), p. 72.

If we let $|\mathcal{Z}(\omega)|$ denote the Lebesgue measure of $\mathcal{Z}(\omega)$, then Fubini's theorem implies

$$E_x(|\mathcal{Z}(\omega)| \cap [0, T]) = \int_0^T P_x(B_t = 0) dt = 0$$

So $\mathcal{Z}(\omega)$ is a set of measure zero.

The last four observations show that $\mathcal{Z}$ is like the Cantor set that is obtained by removing $(1/3, 2/3)$ from $[0, 1]$ and then repeatedly removing the middle third from the intervals that remain. The Cantor set is bigger, however. Its Hausdorff dimension is $\log 2 / \log 3$, while $\mathcal{Z}$ has dimension $1/2$.

7.4.2 Hitting Times

Theorem 7.4.1 *Under P_0 , $\{T_a, a \geq 0\}$ has stationary independent increments.*

Proof The first step is to notice that if $0 < a < b$, then

$$T_b \circ \theta_{T_a} = T_b - T_a,$$

so if f is bounded and measurable, the strong Markov property, 7.3.9, and translation invariance imply

$$\begin{aligned} E_0(f(T_b - T_a) | \mathcal{F}_{T_a}) &= E_0(f(T_b) \circ \theta_{T_a} | \mathcal{F}_{T_a}) \\ &= E_a f(T_b) = E_0 f(T_{b-a}) \end{aligned}$$

To show that the increments are independent, let $a_0 < a_1 \dots < a_n$, let f_i , $1 \leq i \leq n$ be bounded and measurable, and let $F_i = f_i(T_{a_i} - T_{a_{i-1}})$. Conditioning on $\mathcal{F}_{T_{a_{n-1}}}$ and using the preceding calculation, we have

$$E_0\left(\prod_{i=1}^n F_i\right) = E_0\left(\prod_{i=1}^{n-1} F_i \cdot E_0(F_n | \mathcal{F}_{T_{a_{n-1}}})\right) = E_0\left(\prod_{i=1}^{n-1} F_i\right) E_0 F_n$$

By induction, it follows that $E_0 \prod_{i=1}^n F_i = \prod_{i=1}^n E_0 F_i$, which implies the desired conclusion. $\square$

The scaling relation (7.1.1) implies

$$T_a \stackrel{d}{=} a^2 T_1 \tag{7.4.1}$$

Combining Theorem 7.4.1 and (7.4.1), we see that $t_k = T_k - T_{k-1}$ are i.i.d. and

$$\frac{t_1 + \dots + t_n}{n^2} \rightarrow T_1$$

so using Theorem 3.8.8, we see that T_a has a stable law. Since we are dividing by n^2 and $T_a \geq 0$, the index $\alpha = 1/2$ and the skewness parameter $\kappa = 1$, see (3.8.11).

Without knowing the theory mentioned in the previous paragraph, it is easy to determine the Laplace transform

$$\varphi_a(\lambda) = E_0 \exp(-\lambda T_a) \quad \text{for } a \geq 0$$

and reach the same conclusion. To do this, we start by observing that Theorem 7.4.1 implies

$$\varphi_x(\lambda)\varphi_y(\lambda) = \varphi_{x+y}(\lambda).$$

It follows easily from this that

$$\varphi_a(\lambda) = \exp(-ac(\lambda)) \quad (7.4.2)$$

Proof Let $c(\lambda) = -\log \varphi_1(\lambda)$, so (7.4.2) holds when $a = 1$. Using the previous identity with $x = y = 2^{-m}$ and induction gives the result for $a = 2^{-m}$, $m \geq 1$. Then, letting $x = k2^{-m}$ and $y = 2^{-m}$, we get the result for $a = (k+1)2^{-m}$ with $k \geq 1$. Finally, to extend to $a \in [0, \infty)$, note that $a \rightarrow \phi_a(\lambda)$ is decreasing. $\square$

To identify $c(\lambda)$, we observe that (7.4.1) implies

$$E \exp(-T_a) = E \exp(-a^2 T_1)$$

so $ac(1) = c(a^2)$, i.e., $c(\lambda) = c(1)\sqrt{\lambda}$. Since all of our arguments also apply to σB_t we cannot hope to compute $c(1)$. Theorem 7.5.7 will show

$$E_0(\exp(-\lambda T_a)) = \exp(-a\sqrt{2\lambda}) \quad (7.4.3)$$

Our next goal is to compute the distribution of the hitting times T_a . This application of the strong Markov property shows why we want to allow the function Y that we apply to the shifted path to depend on the stopping time S .

Example 7.4.2 (Reflection principle) Let $a > 0$ and let $T_a = \inf\{t : B_t = a\}$. Then

$$P_0(T_a < t) = 2P_0(B_t \geq a) \quad (7.4.4)$$

Intuitive proof. We observe that if B_s hits a at some time $s < t$, then the strong Markov property implies that $B_t - B(T_a)$ is independent of what happened before time T_a . The symmetry of the normal distribution and $P_a(B_u = a) = 0$ for $u > 0$ then imply

$$P_0(T_a < t, B_t > a) = \frac{1}{2}P_0(T_a < t) \quad (7.4.5)$$

Rearranging the last equation and using $\{B_t > a\} \subset \{T_a < t\}$ gives

$$P_0(T_a < t) = 2P_0(T_a < t, B_t > a) = 2P_0(B_t > a)$$

Proof To make the intuitive proof rigorous, we only have to prove (7.4.5). To extract this from the strong Markov property, Theorem 7.3.9, we let

$$Y_s(\omega) = \begin{cases} 1 & \text{if } s < t, \omega(t-s) > a \\ 0 & \text{otherwise} \end{cases}$$

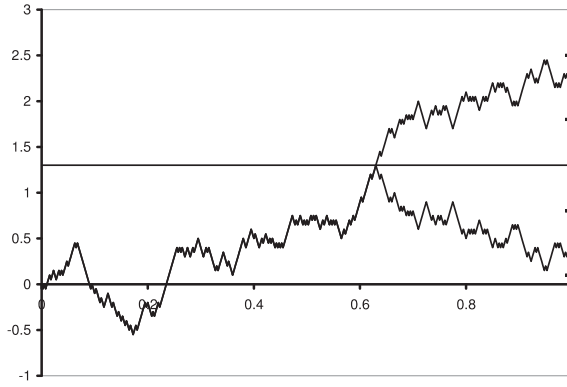


Figure 7.3 Proof by picture of the reflection principle.

We do this so that if we let $S = \inf\{s < t : B_s = a\}$ with $\inf \emptyset = \infty$, then

$$Y_S(\theta_S \omega) = \begin{cases} 1 & \text{if } S < t, B_t > a \\ 0 & \text{otherwise} \end{cases}$$

and the strong Markov property implies

$$E_0(Y_S \circ \theta_S | \mathcal{F}_S) = \varphi(B_S, S) \quad \text{on } \{S < \infty\} = \{T_a < t\}$$

where $\varphi(x, s) = E_x Y_s$. $B_S = a$ on $\{S < \infty\}$ and $\varphi(a, s) = 1/2$ if $s < t$, so taking expected values gives

$$\begin{aligned} P_0(T_a < t, B_t \geq a) &= E_0(Y_S \circ \theta_S; S < \infty) \\ &= E_0(E_0(Y_S \circ \theta_S | \mathcal{F}_S); S < \infty) = E_0(1/2; T_a < t) \end{aligned}$$

which proves (7.4.5). □

Using (7.4.4), we can compute the probability density of T_a . We begin by noting that

$$P(T_a \leq t) = 2 P_0(B_t \geq a) = 2 \int_a^\infty (2\pi t)^{-1/2} \exp(-x^2/2t) dx$$

then change variables $x = (t^{1/2}a)/s^{1/2}$ to get

$$\begin{aligned} P_0(T_a \leq t) &= 2 \int_t^0 (2\pi t)^{-1/2} \exp(-a^2/2s) \left(-t^{1/2}a/2s^{3/2}\right) ds \\ &= \int_0^t (2\pi s^3)^{-1/2} a \exp(-a^2/2s) ds \end{aligned} \tag{7.4.6}$$

Using the last formula, we can compute:

Example 7.4.3 The distribution of $L = \sup\{t \leq 1 : B_t = 0\}$. By (7.2.4),

$$\begin{aligned}
 P_0(L \leq s) &= \int_{-\infty}^{\infty} p_s(0, x) P_x(T_0 > 1 - s) dx \\
 &= 2 \int_0^{\infty} (2\pi s)^{-1/2} \exp(-x^2/2s) \int_{1-s}^{\infty} (2\pi r^3)^{-1/2} x \exp(-x^2/2r) dr dx \\
 &= \frac{1}{\pi} \int_{1-s}^{\infty} (sr^3)^{-1/2} \int_0^{\infty} x \exp(-x^2(r+s)/2rs) dx dr \\
 &= \frac{1}{\pi} \int_{1-s}^{\infty} (sr^3)^{-1/2} rs/(r+s) dr
 \end{aligned}$$

Our next step is to let $t = s/(r+s)$ to convert the integral over $r \in [1-s, \infty)$ into one over $t \in [0, s]$. $dt = -s/(r+s)^2 dr$, so to make the calculations easier we first rewrite the integral as

$$= \frac{1}{\pi} \int_{1-s}^{\infty} \left(\frac{(r+s)^2}{rs} \right)^{1/2} \frac{s}{(r+s)^2} dr$$

and then change variables to get

$$P_0(L \leq s) = \frac{1}{\pi} \int_0^s (t(1-t))^{-1/2} dt = \frac{2}{\pi} \arcsin(\sqrt{s}) \quad (7.4.7)$$

Since L is the time of the last zero before time 1, it is remarkable that the density is symmetric about $1/2$ and blows up at 0. The arcsin may remind the reader of the limit theorem for $L_{2n} = \sup\{m \leq 2n : S_m = 0\}$ given in Theorem 4.9.5. We will see in Section 8.6 that our new result is a consequence of the old one.

Exercises

7.4.1 Let $B_t = (B_t^1, B_t^2)$ be a two-dimensional Brownian motion starting from 0. Let $T_a = \inf\{t : B_t^2 = a\}$. Show that (i) $B^1(T_a)$, $a \geq 0$ has independent increments and $B^1(T_a) =_d a B^1(T_1)$. Use this to conclude that $B^1(T_a)$ has a Cauchy distribution.

7.4.2 Use (7.2.3) to show that $R = \inf\{t > 1 : B_t = 0\}$ has probability density

$$P_0(R = 1+t) = 1/(\pi t^{1/2}(1+t))$$

7.4.3 (a) Generalize the proof of (7.4.5) to conclude that if $u < v \leq a$, then

$$P_0(T_a < t, u < B_t < v) = P_0(2a - v < B_t < 2a - u) \quad (7.4.8)$$

(b) Let (u, v) shrink down to x in (7.4.8) to conclude that for $a < x$

$$P_0(T_a < t, B_t = x) = p_t(0, 2a - x)$$

i.e., the (subprobability) density for B_t on $\{T_a < t\}$. (c) Use the fact that $\{T_a < t\} = \{M_t > a\}$ and differentiate with respect to a gives the joint density

$$f_{(M_t, B_t)}(a, x) = \frac{2(2a - x)}{\sqrt{2\pi t^3}} e^{-(2a-x)^2/2t}$$

7.4.4 Let $A_{s,t}$ be the event Brownian motion has at least one zero in $[s, t]$. Show that $P_0(A_{s,t}) = (2/\pi)\arccos(\sqrt{s/t})$. Hint: To get started note that

$$P_0(A_{s,t}) = 2 \int_0^\infty (2\pi s)^{-1/2} e^{-x^2/2} P_x(T_0 \leq t-s) dx$$

and use the formula for the density function of T_x . At the end, you may need some trig to convert arctan into arccos.

Remark Using the previous exercise, it is easy to see that if $s \geq 1$ then $P(A_{s,s+h}) \leq Ch^{1/2}$. Thus if we divide the interval $[1, 2]$ into n equal pieces, then expected number of intervals that contains zeros is $\leq Cn^{-1/2}$. It follows that the Hausdorff dimension of the zero set is at most $1/2$.

7.5 Martingales

At the end of Section 5.7, we used martingales to study the hitting times of random walks. The same methods can be used on Brownian motion once we prove:

Theorem 7.5.1 *Let X_t be a right continuous martingale adapted to a right continuous filtration. If T is a bounded stopping time, then $EX_T = EX_0$.*

Proof Let n be an integer so that $P(T \leq n-1) = 1$. As in the proof of the strong Markov property, let $T_m = ([2^m T] + 1)/2^m$. $Y_k^m = X(k2^{-m})$ is a martingale with respect to $\mathcal{F}_k^m = \mathcal{F}(k2^{-m})$ and $S_m = 2^m T_m$ is a stopping time for $(Y_k^m, \mathcal{F}_k^m)$, so by Exercise 4.4.4

$$X(T_m) = Y_{S_m}^m = E(Y_{n2^m}^m | \mathcal{F}_{S_m}^m) = E(X_n | \mathcal{F}(T_m))$$

As $m \uparrow \infty$, $X(T_m) \rightarrow X(T)$ by right continuity and $\mathcal{F}(T_m) \downarrow \mathcal{F}(T)$ by Theorem 7.3.7, so it follows from Theorem 4.7.3 that

$$X(T) = E(X_n | \mathcal{F}(T))$$

Taking expected values gives $EX(T) = EX_n = EX_0$, since X_n is a martingale. □

Theorem 7.5.2 *B_t is a martingale w.r.t. the σ -fields $\mathcal{F}_t$ defined in Section 8.2.*

Note: We will use these σ -fields in all of the martingale results but will not mention them explicitly in the statements.

Proof The Markov property implies that

$$E_x(B_t | \mathcal{F}_s) = E_{B_s}(B_{t-s}) = B_s$$

since symmetry implies $E_y B_u = y$ for all $u \geq 0$. □

From Theorem 7.5.2, it follows immediately that we have:

Theorem 7.5.3 *If $a < x < b$, then $P_x(T_a < T_b) = (b-x)/(b-a)$.*

Proof Let $T = T_a \wedge T_b$. Theorem 7.2.8 implies that $T < \infty$ a.s. Using Theorems 7.5.1 and 7.5.2, it follows that $x = E_x B(T \wedge t)$. Letting $t \rightarrow \infty$ and using the bounded convergence theorem, it follows that

$$x = aP_x(T_a < T_b) + b(1 - P_x(T_a < T_b))$$

Solving for $P_x(T_a < T_b)$ now gives the desired result. □

Theorem 7.5.4 $B_t^2 - t$ is a martingale.

Proof Writing $B_t^2 = (B_s + B_t - B_s)^2$, we have

$$\begin{aligned} E_x(B_t^2 | \mathcal{F}_s) &= E_x(B_s^2 + 2B_s(B_t - B_s) + (B_t - B_s)^2 | \mathcal{F}_s) \\ &= B_s^2 + 2B_s E_x(B_t - B_s | \mathcal{F}_s) + E_x((B_t - B_s)^2 | \mathcal{F}_s) \\ &= B_s^2 + 0 + (t - s) \end{aligned}$$

since $B_t - B_s$ is independent of $\mathcal{F}_s$ and has mean 0 and variance $t - s$. $\square$

Theorem 7.5.5 Let $T = \inf\{t : B_t \notin (a, b)\}$, where $a < 0 < b$.

$$E_0 T = -ab$$

Proof Theorem 7.5.1 and 7.5.4 imply $E_0(B^2(T \wedge t)) = E_0(T \wedge t)$. Letting $t \rightarrow \infty$ and using the monotone convergence theorem gives $E_0(T \wedge t) \uparrow E_0 T$. Using the bounded convergence theorem and Theorem 7.5.3, we have

$$E_0 B^2(T \wedge t) \rightarrow E_0 B_T^2 = a^2 \frac{b}{b-a} + b^2 \frac{-a}{b-a} = ab \frac{a-b}{b-a} = -ab \quad \square$$

Theorem 7.5.6 $\exp(\theta B_t - (\theta^2 t/2))$ is a martingale.

Proof Bringing $\exp(\theta B_s)$ outside

$$\begin{aligned} E_x(\exp(\theta B_t) | \mathcal{F}_s) &= \exp(\theta B_s) E(\exp(\theta(B_t - B_s)) | \mathcal{F}_s) \\ &= \exp(\theta B_s) \exp(\theta^2(t-s)/2) \end{aligned}$$

since $B_t - B_s$ is independent of $\mathcal{F}_s$ and has a normal distribution with mean 0 and variance $t - s$. $\square$

Theorem 7.5.7 If $T_a = \inf\{t : B_t = a\}$, then $E_0 \exp(-\lambda T_a) = \exp(-a\sqrt{2\lambda})$.

Proof Theorem 7.5.1 and 7.5.6 imply that $1 = E_0 \exp(\theta B(T \wedge t) - \theta^2(T \wedge t)/2)$. Taking $\theta = \sqrt{2\lambda}$, letting $t \rightarrow \infty$ and using the bounded convergence theorem gives $1 = E_0 \exp(a\sqrt{2\lambda} - \lambda T_a)$. $\square$

Theorem 7.5.8 If $u(t, x)$ is a polynomial in t and x with

$$\frac{\partial u}{\partial t} + \frac{1}{2} \frac{\partial^2 u}{\partial x^2} = 0 \quad (7.5.1)$$

then $u(t, B_t)$ is a martingale.

Proof Let $p_t(x, y) = (2\pi)^{-1/2} t^{-1/2} \exp(-(y-x)^2/2t)$. The first step is to check that p_t satisfies the heat equation: $\partial p_t / \partial t = (1/2) \partial^2 p_t / \partial y^2$.

$$\begin{aligned} \frac{\partial p}{\partial t} &= -\frac{1}{2} t^{-1} p_t + \frac{(y-x)^2}{2t^2} p_t \\ \frac{\partial p}{\partial y} &= -\frac{y-x}{t} p_t \\ \frac{\partial^2 p}{\partial y^2} &= -\frac{1}{t} p_t + \frac{(y-x)^2}{t^2} p_t \end{aligned}$$

Interchanging $\partial/\partial t$ and $\int$, and using the heat equation

$$\begin{aligned}\frac{\partial}{\partial t} E_x u(t, B_t) &= \int \frac{\partial}{\partial t} (p_t(x, y) u(t, y)) dy \\ &= \int \frac{1}{2} \frac{\partial}{\partial y^2} p_t(x, y) u(t, y) + p_t(x, y) \frac{\partial}{\partial t} u(t, y) dy\end{aligned}$$

Integrating by parts twice the previous

$$= \int p_t(x, y) \left(\frac{\partial}{\partial t} + \frac{1}{2} \frac{\partial}{\partial y^2} \right) u(t, y) dy = 0$$

Since $u(t, y)$ is a polynomial, there is no question about the convergence of integrals and there is no contribution from the boundary terms when we integrate by parts.

At this point, we have shown that $t \rightarrow E_x u(t, B_t)$ is constant. To check the martingale property, we need to show that

$$E(u(t, B_t) | \mathcal{F}_s) = u(s, B_s)$$

To do this, we note that $v(r, x) = u(s + r, x)$ satisfies $\partial v / \partial r = (1/2) \partial^2 v / \partial x^2$, so using the Markov property

$$\begin{aligned}E(u(t, B_t) | \mathcal{F}_s) &= E(v(t - s, B_{t-s}) \circ \theta_s | \mathcal{F}_s) \\ &= E_{B(s)} v(t - s, B_{t-s}) = v(0, B_s) = u(s, B_s)\end{aligned}$$

where in the next to last step we have used the previously proved constancy of the expected value. $\square$

Examples of polynomials that satisfy (7.5.1) are $x, x^2 - t, x^3 - 3tx, x^4 - 6x^2t + 3t^2 \dots$. The result can also be extended to $\exp(\theta x - \theta^2 t / 2)$. The only place we used the fact that $u(t, x)$ was a polynomial was in checking the convergence of the integral to justify integration by parts.

Theorem 7.5.9 *If $T = \inf\{t : B_t \notin (-a, a)\}$, then $ET^2 = 5a^4/3$.*

Proof Theorem 7.5.1 implies

$$E(B(T \wedge t)^4 - 6(T \wedge t)B(T \wedge t)^2) = -3E(T \wedge t)^2.$$

From Theorem 7.5.5, we know that $ET = a^2 < \infty$. Letting $t \rightarrow \infty$, using the dominated convergence theorem on the left-hand side, and the monotone convergence theorem on the right gives

$$a^4 - 6a^2 ET = -3E(T^2)$$

Plugging in $ET = a^2$ gives the desired result. $\square$

Exercises

7.5.1 Let $T = \inf\{B_t \notin (-a, a)\}$. Show that

$$E_0 \exp(-\lambda T) = 1 / \cosh(a\sqrt{2\lambda}).$$

7.5.2 The point of this exercise is to get information about the amount of time it takes Brownian motion with drift $-b$, $X_t \equiv B_t - bt$ to hit level a . Let $\tau = \inf\{t : B_t = a + bt\}$,

where $a > 0$. (i) Use the martingale $\exp(\theta B_t - \theta^2 t/2)$ with $\theta = b + (b^2 + 2\lambda)^{1/2}$ to show

$$E_0 \exp(-\lambda \tau) = \exp(-a\{b + (b^2 + 2\lambda)^{1/2}\})$$

Letting $\lambda \rightarrow 0$ gives $P_0(\tau < \infty) = \exp(-2ab)$.

7.5.3 Let $\sigma = \inf\{t : B_t \notin (a, b)\}$ and let $\lambda > 0$. Use the strong Markov property to show

$$E_x \exp(-\lambda T_a) = E_x(e^{-\lambda \sigma}; T_a < T_b) + E_x(e^{-\lambda \sigma}; T_b < T_a) E_b \exp(-\lambda T_a)$$

(ii) Interchange the roles of a and b to get a second equation, use Theorem 7.5.7, and solve to get

$$E_x(e^{-\lambda \sigma}; T_a < T_b) = \sinh(\sqrt{2\lambda}(b-x)) / \sinh(\sqrt{2\lambda}(b-a))$$

$$E_x(e^{-\lambda \sigma}; T_b < T_a) = \sinh(\sqrt{2\lambda}(x-a)) / \sinh(\sqrt{2\lambda}(b-a))$$

7.5.4 If $T = \inf\{t : B_t \notin (a, b)\}$, where $a < 0 < b$ and $a \neq -b$, then T and B_T^2 are not independent, so we cannot calculate ET^2 as we did in the proof of Theorem 7.5.9. Use the Cauchy-Schwarz inequality to estimate $E(TB_T^2)$ and conclude (i) $ET^2 \leq 4 E(B_T^4)$ and (ii) $EB_T^4 \leq 36ET^2$.

7.5.5 Find a martingale of the form $B_t^6 - c_1 t B_t^4 + c_2 t^2 B_t^2 - c_3 t^3$ and use it to compute the third moment of $T = \inf\{t : B_t \notin (-a, a)\}$.

7.5.6 Show that $(1+t)^{-1/2} \exp(B_t^2/2(1+t))$ is a martingale and use this to conclude that $\limsup_{t \rightarrow \infty} B_t / ((1+t) \log(1+t))^{1/2} \leq 1$ a.s.

7.6 Itô's Formula*

Our goal in this section is to prove three versions of Itô's formula. The first one, for one-dimensional Brownian motion is:

Theorem 7.6.1 Suppose that $f \in C^2$, i.e., it has two continuous derivatives. With probability one, for all $t \geq 0$,

$$f(B_t) - f(B_0) = \int_0^t f'(B_s) dB_s + \frac{1}{2} \int_0^t f''(B_s) ds \quad (7.6.1)$$

Of course, as part of the derivation we have to explain the meaning of the first term on the right-hand side. In contrast, if A_s is continuous and has bounded variation on each finite interval, and $f \in C^1$:

$$f(A_t) - f(A_0) = \int_0^t f'(A_s) dA_s$$

where the Riemann-Stieltjes integral on the right-hand side is

$$\lim_{n \rightarrow \infty} \sum_{i=1}^{k(n)} f'(A_{t_{i-1}^n}) [A(t_i^n) - A(t_{i-1}^n)]$$

over a sequence of partitions $0 = t_0^n < t_1^n < \dots < t_{k(n)}^n = t$ with mesh $\max_i t_i^n - t_{i-1}^n \rightarrow 0$.

Proof To derive (7.6.1), we let $t_i^n = ti/2^n$ for $0 \leq i \leq k(n) = 2^n$. From calculus we know that for any a and b there is a $c(a, b)$ in between a and b such that

$$f(b) - f(a) = (b - a)f'(a) + \frac{1}{2}(b - a)^2 f''(c(a, b)) \quad (7.6.2)$$

Using the calculus fact, it follows that

$$\begin{aligned} f(B_t) - f(B_0) &= \sum_i f(B_{t_{i+1}^n}) - f(B_{t_i^n}) \\ &= \sum_i f'(B_{t_i^n})(B_{t_{i+1}^n} - B_{t_i^n}) \\ &\quad + \frac{1}{2} \sum_i g_i^n(\omega)(B_{t_{i+1}^n} - B_{t_i^n})^2 \end{aligned} \quad (7.6.3)$$

where $g_i^n(\omega) = f''(c(B_{t_i^n}, B_{t_{i+1}^n}))$. Comparing (7.6.3) with (7.6.1), it becomes clear that we want to show

$$\sum_i f'(B_{t_i^n})(B_{t_{i+1}^n} - B_{t_i^n}) \rightarrow \int_0^t f'(B_s) dB_s \quad (7.6.4)$$

$$\frac{1}{2} \sum_i g_i^n(\omega)(B_{t_{i+1}^n} - B_{t_i^n})^2 \rightarrow \frac{1}{2} \int_0^t f''(B_s) ds \quad (7.6.5)$$

in probability as $n \rightarrow \infty$.

We will prove the result first under the assumption that $|f'(x)|, |f''(x)| \leq K$. Let

$$I_m^1 \equiv \sum_i f'(B_{t_i^m})(B_{t_{i+1}^m} - B_{t_i^m})$$

In order to compare I_m^1 and I_n^1 with $n \geq m$, we want to further subdivide the intervals of length $t2^{-m}$ to be of length $t2^{-n}$. To leave the sum unchanged, we have to redefine things. Let $i(n, m, j) = [2^m j / 2^n] / 2^m$ and $H_j^m = f'(B_{i(n, m, j)})$. If we let $K_j^n = f'(B_{t_j^n})$, then we have

$$I_m^1 - I_n^1 = \sum_j (H_j^m - K_j^n)(B_{t_{j+1}^n} - B_{t_j^n})$$

Squaring, we have

$$\begin{aligned} (I_m^1 - I_n^1)^2 &= \sum_j (H_j^m - K_j^n)^2 (B_{t_{j+1}^n} - B_{t_j^n})^2 \\ &\quad + 2 \sum_{j < k} (H_j^m - K_j^n)(H_k^m - K_k^n)(B_{t_{j+1}^n} - B_{t_j^n})(B_{t_{k+1}^n} - B_{t_k^n}) \end{aligned}$$

The next step is to take expected value. The expected value of the second sum vanishes because if we take expected value of the j, k term with respect to $\mathcal{F}_{t_k^n}$, the first three terms can be taken outside the expected value

$$(H_j^m - K_j^n)(H_k^m - K_k^n)(B_{t_{j+1}^n} - B_{t_j^n})E(B_{t_{k+1}^n} - B_{t_k^n} | \mathcal{F}_{t_k^n}) = 0$$

Conditioning on $\mathcal{F}_{t_j^n}$, in the first sum we have

$$E(I_m^1 - I_n^1)^2 = \sum_j E(H_j^m - K_j^n)^2 \cdot 2^{-n}t \quad (7.6.6)$$

Since $|i(n, m, j)/2^m - j/2^n| \leq 1/2^m$ and $|f''(x)| \leq K$, we have

$$\sup_{j \leq 2^n} |H_j^m - K_j^n| \leq K \sup\{|B_s - B_r| : 0 \leq r \leq s \leq t, |s - r| \leq 2^{-m}\}$$

Since Brownian motion is uniformly continuous on $[0, t]$ and $E \sup_{s \leq t} |B_s|^2 < \infty$, it follows from the dominated convergence theorem that

$$\lim_{m, n \rightarrow \infty} E(I_m^1 - I_n^1)^2 = 0$$

This shows that I_m^1 is a Cauchy sequence in L^2 , so there is an I_∞^1 so that $I_m^1 \rightarrow I_\infty^1$ in L^2 . The limit defines $\int_0^t f'(B_s) dB_s$ as a limit of approximating Riemann sums.

To prove (7.6.5), we let $G_s^n = g_i^n(\omega) = f''(c(B_{t_i^n}, B_{t_{i+1}^n}))$ when $s \in (t_i^n, t_{i+1}^n]$ and let

$$A_s^n = \sum_{t_{i+1}^n \leq s} (B_{t_{i+1}^n} - B_{t_i^n})^2$$

so that

$$\sum_i g_i^n(\omega) (B_{t_{i+1}^n} - B_{t_i^n})^2 = \int_0^t G_s^n dA_s^n$$

and what we want to show is

$$\int_0^t G_s^n dA_s^n \rightarrow \int_0^t f''(B_s) ds \quad (7.6.7)$$

To do this, we begin by observing that continuity of f'' implies that if $s_n \rightarrow s$ as $n \rightarrow \infty$, then $G_{s_n}^n \rightarrow f''(B_s)$, while Exercise 7.1.6 implies that A_s^n converges almost surely to s . At this point, we can fix ω and deduce (7.6.7) from the following simple result:

Lemma 7.6.2 *If (i) measures μ_n on $[0, t]$ converge weakly to μ_∞ , a finite measure, and (ii) g_n is a sequence of functions with $|g_n| \leq K$ that have the property that whenever $s_n \in [0, t] \rightarrow s$ we have $g_n(s_n) \rightarrow g(s)$, then as $n \rightarrow \infty$*

$$\int g_n d\mu_n \rightarrow \int g d\mu_\infty$$

Proof By letting $\mu'_n(A) = \mu_n(A)/\mu_n([0, t])$, we can assume that all the μ_n are probability measures. A standard construction (see Theorem 3.2.8) shows that there is a sequence of random variables X_n with distribution μ_n so that $X_n \rightarrow X_\infty$ a.s. as $n \rightarrow \infty$. The convergence of g_n to g implies $g_n(X_n) \rightarrow g(X_\infty)$, so the result follows from the bounded convergence theorem. $\square$

Lemma 7.6.2 is the last piece in the proof of Theorem under the additional assumptions that $|f'(x)|, |f''(x)| \leq M$. Tracing back through the proof, we see that Lemma 7.6.2 implies

(7.6.7), which in turn completes the proof of (7.6.5). So adding (7.6.4) and using (7.6.3) gives that

$$f(B_t) - f(B_0) = \int_0^t f'(B_s) dB_s + \frac{1}{2} \int_0^t f''(B_s) ds \quad \text{a.s.}$$

To remove the assumed bounds on the derivatives, we let M be large and define f_M with $f_M = f$ on $[-M, M]$ and $|f'_M|, |f''_M| \leq M$. The formula holds for f_M but on $\{\sup_{s \leq t} |B_s| \leq M\}$ there is no difference in the formula between using f_M and f . $\square$

Example 7.6.3 Taking $f = x^2$ in (7.6.1), we have

$$B_t^2 - B_0^2 = \int_0^t 2B_s dB_s + t$$

so if $B_0 = 0$, we have

$$\int_0^t 2B_s dB_s = B_t^2 - t$$

In contrast to the calculus formula $\int_0^a 2x dx = a^2$.

Direct proof. Using the subdivision we introduced previously, we can write

$$\begin{aligned} B_t^2 &= \left(\sum_{i=0}^{2^n-1} (B(t_{i+1}^n) - B(t_i^n)) \right)^2 \\ &\quad + \sum_{i=0}^{2^n-1} \sum_{j=0}^{2^n-1} (B(t_{i+1}^n) - B(t_i^n))(B(t_{j+1}^n) - B(t_j^n)) \end{aligned}$$

If we sum over $j < i$, this becomes

$$\sum_{i=0}^{2^n-1} B(t_i^n - i)(B(t_{i+1}^n) - B(t_i^n)) \rightarrow \int_0^t B_s dB_s$$

We get the same limit for the sum over $i < j$. When we sum over $i = j$

$$\sum_{i=0}^{2^n-1} (B(t_{i+1}^n) - B(t_i^n))^2 \rightarrow t$$

To give one reason why the answer cannot be $B_t^2/2$, we note that:

Theorem 7.6.4 *If g is continuous and $E \int_0^t |g(B_s)|^2 ds < \infty$, then $\int_0^t g(B_s) dB_s$ is a continuous martingale.*

If g grows too fast at ∞ , then the integrability condition might not hold. To deal with this, we introduce a weaker concept: M_t is a local martingale if there is an increasing sequence of stopping times $T_n \uparrow \infty$ so that $M(t \wedge T_n)$ is a martingale. If g is continuous, then the stochastic integral is always a local martingale.

Proof We first prove the result assuming $|g| \leq K$. Let $t_i^n = ti/2^n$ for $0 \leq i \leq 2^n$, let $J = \max\{i : t_{i+1}^n \leq s\}$, and let

$$I_n^1(s) = \sum_{i=0}^J g(B_{t_i^n})(B_{t_{i+1}^n} - B_{t_i^n}) + g(B_{t_{J+1}^n})(B_s - B_{t_{J+1}^n})$$

It is easy to check that $I_n^1(s)$, $s \leq t$ is a continuous martingale with

$$EI_n^1(t)^2 = \sum_{0 \leq i < 2^n} E[g(B_{t_i^n})^2] \cdot t2^{-n}.$$

The L^2 maximal inequality implies that

$$E \left(\sup_{s \leq t} |I_m^1(s) - I_n^1(s)|^2 \right) \leq 4E(I_m^1 - I_n^1)^2$$

The right-hand side $\rightarrow 0$ by the previous proof, so $I_m^1(s) \rightarrow I_\infty^1(s)$ uniformly on $[0, t]$. Let $Y \in \mathcal{F}_r$ with $EY^2 < \infty$. Since $I_n^1(s)$ is a martingale, if $r < s < t$, we have

$$E[Y\{I_n^1(s) - I_n^1(r)\}] = 0$$

From this and the Cauchy-Schwarz inequality we conclude that

$$E[Y\{I_\infty^1(s) - I_\infty^1(r)\}] = 0$$

Since this holds for $Y = 1_A$ with $A \in \mathcal{F}_r$, we have proved the desired result under the assumption $|g| \leq K$.

Approximating g by $g_M = g$ on $[-M, M]$ and g_M bounded as before and stopping at time $T_M = \inf\{t : |B_t| \geq M\}$, we can conclude that if g is continuous, then

$$\int_0^{s \wedge T_M} g(B_r) dB_r$$

is a continuous martingale with

$$E \left(\int_0^{s \wedge T_M} f'(B_r) dB_r \right)^2 = E \int_0^{s \wedge T_M} f'(B_r)^2 dr$$

Letting $M \rightarrow \infty$ it follows that if $E \int_0^t |f'(B_s)|^2 ds < \infty$, then $\int_0^t f'(B_s) dB_s$ is a continuous martingale. $\square$

Our second version of Itô's formula allows f to depend on t as well.

Theorem 7.6.5 Suppose $f \in C^2$, i.e., it has continuous partial derivatives up to order two. Then with probability 1, for all $t \geq 0$

$$\begin{aligned} f(t, B_t) - f(0, B_0) &= \int_0^t \frac{\partial f}{\partial t}(s, B_s) ds + \int_0^t \frac{\partial f}{\partial x}(s, B_s) dB_s \\ &\quad + \frac{1}{2} \int_0^t \frac{\partial^2 f}{\partial x^2}(s, B_s) ds \end{aligned} \quad (7.6.8)$$

Proof Perhaps the simplest way to give a completely rigorous proof of this result (and the next one) is to prove the result when f is a polynomial in t and x and then note that for any $f \in C^2$ and $M < \infty$, we can find polynomials ϕ_n so that ϕ_n and all of its partial derivatives up to order two converge to those of f uniformly on $[0, t] \times [-M, M]$. Here, we will content ourselves to explain why this result is true.

Expanding to second order and being a little less precise about the error term

$$\begin{aligned} f(t, b) - f(s, a) &\approx \frac{\partial f}{\partial t}(s, a)(t - s) + \frac{\partial f}{\partial x}(s, a)(b - a) \\ &+ \frac{1}{2} \frac{\partial^2 f}{\partial x^2}(s, a)(b - a)^2 + \frac{\partial^2 f}{\partial t \partial x}(s, a)(t - s)(b - a) + \frac{1}{2} \frac{\partial^2 f}{\partial t^2}(s, a)(t - s)^2 \end{aligned}$$

Let t be a fixed positive number and let $t_k^n = tk/2^n$. As before, we first prove the result under the assumption that all of the first- and second-order partial derivatives are bounded by K . The details of the convergence of the first three terms to their limits is similar to the previous proof. To see that the fifth term vanishes in the limit, use the fact that $t_{i+1}^n - t_i^n = 2^{-n}$ to conclude that

$$\sum_i \left| \frac{\partial^2 f}{\partial x^2}(t_i^n, B(t_i^n)) \right| (t_{i+1}^n - t_i^n)^2 \leq Kt2^{-n} \rightarrow 0$$

To treat the fourth term, we first consider

$$V_n^{1,1}(t) = \sum_i (t_{i+1}^n - t_i^n) |B(t_{i+1}^n) - B(t_i^n)|$$

$EV_n^{1,1}(t) = \sum_i C_1 (t_{i+1}^n - t_i^n)^{3/2}$, where $C_1 = E\chi^{1/2}$ and χ is a standard normal. Arguing as before,

$$E \sum_i \left| \frac{\partial^2 f}{\partial x^2}(t_i^n, B(t_i^n)) \right| (t_{i+1}^n - t_i^n) |B(t_{i+1}^n) - B(t_i^n)| \leq KC_1 t \cdot 2^{-n/2} \rightarrow 0$$

Using Markov's inequality now, we see that if $\epsilon > 0$

$$P \left(\sum_i \left| \frac{\partial^2 f}{\partial x^2}(t_i^n, B(t_i^n)) \right| (t_{i+1}^n - t_i^n) |B(t_{i+1}^n) - B(t_i^n)| > \epsilon \right) \leq KC_1 t 2^{-n/2} / \epsilon \rightarrow 0$$

and hence the sum converges to 0 in probability.

To remove the assumption that the partial derivatives are bounded, let $T_M = \min\{t : |B_t| > M\}$ and use the previous calculation to conclude that the formula holds up to time $T_M \wedge t$. $\square$

Example 7.6.6 Using (7.6.8), we can prove Theorem 7.5.8: if $u(t, x)$ is a polynomial in t and x with $\partial u / \partial t + (1/2) \partial^2 u / \partial x^2 = 0$, then Itô's formula implies

$$u(t, B_t) - u(0, B_0) = \int_0^t \frac{\partial u}{\partial x}(s, B_s) dB_s$$

Since $\partial u / \partial x$ is a polynomial, it satisfies the integrability condition in Theorem 7.6.4; $u(t, B_t)$ is a martingale. To get a new conclusion, let $u(t, x) = \exp(\mu t + \sigma x)$. Exponential Brownian motion $X_t = u(t, B_t)$ is often used as a model of the price of a stock. Since

$$\frac{\partial u}{\partial t} = \mu u \quad \frac{\partial u}{\partial x} = \sigma u \quad \frac{\partial^2 u}{\partial x^2} = \sigma^2 u$$

Itô's formula gives

$$X_t - X_0 = \int_0^t \left(\mu - \frac{\sigma^2}{2} \right) X_s ds + \int_0^t \sigma X_s dB_s$$

From this we see that if $\mu = \sigma^2/2$, then X_t is a martingale, but this formula gives more information about how X_t behaves.

Our third and final version of Itô's formula is for multi-dimensional Brownian motion. To simplify notation we write $D_i f = \partial f / \partial x_i$ and $D_{ij} = \partial^2 f / \partial x_i \partial x_j$.

Theorem 7.6.7 Suppose $f \in C^2$. Then with probability 1, for all $t \geq 0$

$$\begin{aligned} f(t, B_t) - f(0, B_0) &= \int_0^t \frac{\partial f}{\partial t}(s, B_s) ds + \sum_{i=1}^d \int_0^t D_i f(B_s) dB_s^i \\ &\quad + \frac{1}{2} \sum_{i=1}^d \int_0^t D_{ii} f(B_s) ds \end{aligned} \quad (7.6.9)$$

Proof Let $x_0 = t$ and write $D_0 f = \partial f / \partial x_0$. Expanding to second order and again not worrying about the error term

$$\begin{aligned} f(y) - f(x) &\approx \sum_i D_i f(x)(y_i - x_i) \\ &\quad + \frac{1}{2} \sum_i D_{ii} f(x)(y_i - x_i)^2 + \sum_{i < j} D_{ij} f(x)(y_i - x_i)(y_j - x_j) \end{aligned}$$

Let t be a fixed positive number and let $t_i^n t_i / 2^n$. The details of the convergence of the first two terms to their limits are similar to the previous two proofs. To handle the third term, we let

$$F_{i,j}(t_k^n) = D_{ij} f(B(t_k^n))$$

and note that independence of coordinates implies

$$\begin{aligned} E(F_{i,j}(t_k^n)(B^i(t_{k+1}^n) - B^i(t_k^n)) \cdot (B^j(t_{k+1}^n) - B^j(t_k^n)) | \mathcal{F}_{t_k^n}) \\ = F_{i,j}(t_k^n) E((B^i(t_{k+1}^n) - B^i(t_k^n)) \cdot (B^j(t_{k+1}^n) - B^j(t_k^n)) | \mathcal{F}_{t_k^n}) = 0 \end{aligned}$$

Fix $i < j$ and let $G_k = F_{i,j}(t_k^n)(B^i(t_{k+1}^n) - B^i(t_k^n)) \cdot (B^j(t_{k+1}^n) - B^j(t_k^n))$. The last calculation shows that

$$E \left(\sum_k G_k \right) = 0$$

To compute the variance of the sum, we note that

$$E \left(\sum_k G_k \right)^2 = E \sum_k G_k^2 + 2E \sum_{k < \ell} G_k G_\ell$$

Again we first complete the proof under the assumption that $|\partial^2 f / \partial x_i \partial x_j| \leq C$. In this case, conditioning on $\mathcal{F}_{t_k^n}$ in order to bring $F_{i,j}^2(t_k^n)$ outside the first term,

$$\begin{aligned}
& E(F_{i,j}^2(t_k^n)(B^i(t_{k+1}^n) - B^i(t_k^n))^2 \cdot (B^j(t_{k+1}^n) - B^j(t_k^n))^2 | \mathcal{F}_{t_k^n}) \\
&= F_{i,j}^2(t_k^n) E((B^i(t_{k+1}^n) - B^i(t_k^n))^2 \cdot (B^j(t_{k+1}^n) - B^j(t_k^n))^2 | \mathcal{F}_{t_k^n}) = F_{i,j}^2(t_k^n)(t_{k+1}^n - t_k^n)^2
\end{aligned}$$

and it follows that

$$E \sum_k G_k^2 \leq K^2 \sum_k (t_{k+1}^n - t_k^n)^2 \leq K^2 t 2^{-n} \rightarrow 0$$

To handle the cross terms, we condition on $\mathcal{F}_{t_\ell^n}$ to get

$$E(G_k G_\ell | \mathcal{F}_{t_k^n}) = G_k F_{i,j}(t_\ell^n) E((B^i(t_{\ell+1}^n) - B^i(t_\ell^n))(B^j(t_{\ell+1}^n) - B^j(t_\ell^n)) | \mathcal{F}_{t_\ell^n}) = 0$$

At this point, we have shown $E(\sum_k G_k)^2 \rightarrow 0$ and the desired result follows from Markov's inequality. The assumption that the second derivatives are bounded can be removed as before. $\square$

Exercises

- 7.6.1 Suppose $f \in C^2$. Show that $f(B_t)$ is a martingale if and only if $f(x) = a + bx$.
- 7.6.2 Show that $B_t^3 - 3tB_t$ and $B_t^3 - \int_0^t 3B_s ds$ are a martingale.
- 7.6.3 Let $\beta_{2k}(t) = E_0 B_t^{2k}$. Use Itô's formula to relate $\beta_{2k}(t)$ to $\beta_{2k-2}(t)$ and use this relationship to derive a formula for $\beta_{2k}(t)$.
- 7.6.4 Apply Itô's formula to $|B_t|$.
- 7.6.5 Apply Itô's formula to $|B_t|^2$. Use this to conclude that $E_0 |B_t|^2 = td$.